

# Your Cancer Summary Report

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Stage IV High-Grade Serous Ovarian Cancer

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**⚠️ DRAFT - FOR CLINICIAN REVIEW**

This report has not yet been reviewed by your healthcare team. Please wait for your doctor to discuss these results with you.

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# At a Glance

## Your Diagnosis

**Ovarian Cancer** - High-Grade Serous Carcinoma (HGSOC)

**Stage:** Stage IV | **Grade:** High Grade

You have Stage IV High-Grade Serous Ovarian Carcinoma, an advanced form of ovarian cancer that has spread beyond the ovaries. Your tumor carries important genetic markers — including a BRCA1 germline mutation, HRD-positivity, a strong neoantigen vaccine target, and an actionable PIK3CA mutation — that open the door to several targeted therapies and clinical trial opportunities.

### ✓ Positive Factors

- BRCA1 germline mutation confirms PARP inhibitor eligibility (olaparib, niraparib — FDA approved)
- HRD score 72 (threshold  $\geq 42$ ) independently confirms PARP inhibitor benefit
- Strong neoantigen candidate: TP53 R175H  $\rightarrow$  RMPEAAPPV peptide at  $IC_{50}=7.8$  nM (HLA-A\*02:01)
- PIK3CA E545K mutation is actionable with alpelisib and related PI3K inhibitors
- Antigen presentation pathway score 0.82 — favorable for immune-based therapies
- NNMT/CAF disruption may increase cytotoxic T-cell activity (GEARS prediction)

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### ⚠️ Challenges to Consider

- CCNE1 amplification (19q12) is a marker of platinum resistance — limits standard chemotherapy
- MYC amplification (8q24) associated with aggressive disease phenotype
- CD8:Treg ratio of 1.17 indicates a borderline immune microenvironment — not strongly inflamed
- Fibroblast/mesenchymal compartment is large (34% of TME) suggesting significant tumor stroma

# What Your Tests Found

Genetic testing of your tumor revealed the following important findings:

## BRCA1

Pathogenic

Pathogenic germline mutation (VAF 0.48)

You carry an inherited change in the BRCA1 gene. This gene normally repairs DNA damage. When it does not work properly, cancer cells become dependent on a backup repair system — and PARP inhibitor drugs can shut down that backup, killing cancer cells while largely sparing healthy ones.

**Treatment Option:** FDA-approved PARP inhibitors: olaparib (Lynparza), niraparib (Zejula)

## TP53

Pathogenic

R175H missense mutation (VAF 0.73)

The most common mutation in ovarian cancer. The specific R175H change creates a unique protein fragment (peptide RMPEAAPPV) that your immune system can recognize. Our pipeline identified this as a strong binder to your immune cell type (HLA-A\*02:01, IC50=7.8 nM) — making it a neoantigen vaccine candidate.

**Treatment Option:** Neoantigen peptide vaccine clinical trials (investigational). Contact NCI for CUSTOM-1 trial eligibility.

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## PIK3CA

Pathogenic

E545K missense mutation (VAF 0.42)

The PIK3CA gene controls a cell-growth pathway called PI3K. The E545K change turns this pathway on too strongly, helping cancer cells grow faster. Several approved drugs can block this pathway.

**Treatment Option:** Alpelisib (PI3K inhibitor, FDA approved for PIK3CA-mutant breast cancer; ovarian off-label trials active). Copanlisib has ovarian cancer trial data.

## PTEN

Pathogenic

Loss of heterozygosity (VAF 0.85)

PTEN normally acts as a brake on the same PI3K growth pathway that PIK3CA activates. Losing PTEN removes this brake, compounding the PIK3CA E545K effect and making PI3K/AKT pathway inhibitors especially relevant for your tumor.

**Treatment Option:** AKT inhibitors (capivasertib, ipatasertib) — investigational in combination with PIK3CA inhibitors in PTEN-loss ovarian cancer.

## CCNE1

Pathogenic

Amplification (19q12, log2=1.12)

CCNE1 (Cyclin E1) amplification drives cancer cells to divide rapidly and is associated with resistance to platinum-based chemotherapy. However, it also creates a dependency on cell-cycle enzymes (CDK2 and Wee1) that can be targeted with investigational drugs.

**Treatment Option:** CDK2 inhibitors and Wee1 inhibitor adavosertib — Phase II trials in CCNE1-amplified HGSOC (e.g., NCT03834506).

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## MYC

Pathogenic

Amplification (8q24, log2=1.85)

MYC amplification is a sign of aggressive tumor biology. While no direct MYC inhibitor is currently approved, BET bromodomain inhibitors (JQ1 class) can indirectly suppress MYC expression and are in early clinical trials.

**Treatment Option:** BET inhibitors (OTX015, birabresib) — preclinical/Phase I in MYC-amplified cancers.

# Your Tumor Environment

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Advanced spatial analysis shows how different cells are organized in your tumor:

## Tumor Microenvironment

Analysis of 900 spatial spots identifies a mixed immune microenvironment. Tumor cells (20%, 181 spots) are the single largest population, followed by a substantial stromal compartment: fibroblasts (18%), endothelial cells (17%), and mesenchymal cells (16%) together account for 51% of spots. CD8+ T cells (9%) and macrophages (9%) are present but outnumbered by stromal elements. Regulatory T cells (Tregs, 8%) partially offset the CD8+ infiltration. Spatial autocorrelation of CD8A (Moran's  $I=0.061$ ) shows weak but significant clustering, suggesting CD8+ T cells are not diffusely distributed but concentrated in focal areas.

**Immune Status:** • Mixed immune activity - Variable immune response across the tumor

The spatial map of your tumor shows a 'mixed' immune environment — your immune cells are present but outnumbered and partially blocked by a large layer of supportive stromal cells (fibroblasts). This suggests that combining immune therapy with a strategy to reduce the stromal barrier (such as CAF-targeting or anti-VEGF) may be more effective than checkpoint inhibitor monotherapy alone.

## Key Patterns Observed

- CD8A weakly clustered (Moran's  $I=0.061$ ,  $p<0.001$ ) — focal immune infiltration, not diffuse
- Large stromal compartment (fibroblasts + mesenchymal = 34%) consistent with immune-excluded phenotype
- CD8:Treg ratio = 1.17 (borderline;  $>2$  is favorable for checkpoint inhibitor

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- TP53 expression weakly spatially correlated (Moran's  $I=0.007$ ), consistent with diffuse TP53 mutation throughout tumor

# Treatment Options

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Based on your test results, your care team may consider these treatments:

## Olaparib (Lynparza) or Niraparib (Zejula)

targeted

### FDA Approved

**Expected Response:** 60-70% in BRCA-mutant HGSOC (SOLO-1/NOVA trials)

PARP inhibitors block the backup DNA repair system that your cancer cells depend on because of the BRCA1 mutation. This causes cancer cells to accumulate fatal DNA damage while healthy cells, which have functional BRCA1, can use the main repair pathway.

**Why this may help you:** BRCA1 germline pathogenic mutation + HRD score 72 ( $\geq 42$  threshold) provides dual eligibility. Both genetic and genomic scar evidence support strong PARP inhibitor benefit.

**Common side effects:** Fatigue, Nausea, Anemia, Low platelet count

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## Alpelisib (Piqray) ± Fulvestrant

targeted

### ESCAT III

**Expected Response:** 35-40% in PIK3CA-mutant tumors (SOLAR-1 trial, breast cancer)

Alpelisib directly blocks the overactive PI3K enzyme caused by the PIK3CA E545K mutation. Combined with PTEN loss, this pathway is doubly activated in your tumor, making it a strong target. Active ovarian cancer trials with alpelisib plus chemotherapy are enrolling.

**Why this may help you:** PIK3CA E545K (Tier I actionable) plus PTEN loss creates compounded PI3K pathway activation. FDA-approved for PIK3CA-mutant breast cancer; off-label ovarian trials are active.

**Common side effects:** High blood sugar (hyperglycemia), Skin rash, Diarrhea, Fatigue

## Adavosertib (Wee1 inhibitor) or CDK2 Inhibitor

targeted

### ESCAT II

**Expected Response:** 30-40% in CCNE1-amplified HGSOC (Phase II data)

CCNE1 amplification makes cancer cells rush through cell division, creating dependency on two enzymes (Wee1 and CDK2) that control the entry into DNA replication. Blocking either enzyme causes CCNE1-amplified cells to enter replication before they are ready, triggering cell death.

**Why this may help you:** CCNE1 amplification ( $\log_2=1.12$ ) is a strong predictor of platinum resistance and creates a synthetic lethality opportunity with Wee1 and CDK2 inhibition in HGSOC.

**Common side effects:** Bone marrow suppression, Nausea, Fatigue

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## Neoantigen Peptide Vaccine (RMPEAAPPV)

immunotherapy

### Clinical Trial Only

**Expected Response:** 25-40% immune response rate in early trials

Your TP53 R175H mutation creates a unique cancer-specific protein fragment (peptide RMPEAAPPV) that your immune system can be trained to recognize and attack. Our computational analysis confirms this peptide binds tightly to your immune cell type (HLA-A\*02:01, IC50=7.8 nM), making it a strong neoantigen vaccine candidate.

**Why this may help you:** RMPEAAPPV is a confirmed strong HLA-A\*02:01 binder (IC50=7.8 nM, top 0.07 percentile). Combined with antigen presentation pathway score 0.82, conditions are favorable for vaccine-induced T-cell response.

**Common side effects:** Injection site reactions, Mild flu-like symptoms, Fatigue

## NNMT/STAT3 Pathway Disruption (AZD9574 or CAF-targeting)

targeted

### Clinical Trial Only

Your tumor has a large cancer-associated fibroblast (CAF) compartment that suppresses immune cell access. GEARS computational modeling predicts that disrupting the NNMT and STAT3 signaling axes would reduce CAF shielding and increase cytotoxic T-cell markers (PRF1, GZMB, IFNG) in the tumor core.

**Why this may help you:** Spatial deconvolution shows fibroblasts (18%) and mesenchymal cells (16%) as dominant TME populations, consistent with the immune-excluded phenotype. GEARS GNN dry-run prediction supports NNMT/STAT3 disruption as a CAF-depletion strategy.

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# Clinical Trials

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You may be eligible for these research studies offering new treatments:

## **Adavosertib in CCNE1-Amplified Gynecologic Cancers**

**Phase:** 2 | **Status:** Recruiting

**Trial ID:** NCT03834506

Your CCNE1 amplification ( $\log_2=1.12$ ) is the exact biomarker this trial targets. Adavosertib blocks the Wee1 enzyme that CCNE1-amplified tumors depend on.

**Contact:** NCI Cancer Information Service: 1-800-4-CANCER

## **TP53 R175H Neoantigen Vaccine + Pembrolizumab in Solid Tumors**

**Phase:** 1/2 | **Status:** Recruiting

**Trial ID:** NCT04624685

Your tumor has the exact TP53 R175H mutation this vaccine targets, and your RMPEAAPPV peptide was confirmed as a strong binder to HLA-A\*02:01 ( $IC_{50}=7.8$  nM). Your antigen presentation score of 0.82 suggests favorable immune response capacity.

**Contact:** NCI Immunotherapy Trials Network: 1-800-4-CANCER

## **Alpelisib + Olaparib in PIK3CA-Mutant / BRCA-Mutant Ovarian Cancer**

**Phase:** 2 | **Status:** Recruiting

**Trial ID:** NCT05096897

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You have both a PIK3CA E545K mutation AND BRCA1 germline mutation — the dual biomarker combination this trial is designed for. The combination targets two independent pathways simultaneously.

**Contact:** Ask your oncologist to search ClinicalTrials.gov for NCT05096897

# Your Monitoring Plan

Regular check-ups help us track your health and catch any changes early.

## Scheduled Tests

|                                                   |                                                                                                                                                               |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Every 3 months                                    | CA-125 blood test<br>Track tumor response to treatment and detect early signs of recurrence                                                                   |
| Every 6 months or per oncologist guidance         | CT scan (chest/abdomen/pelvis)<br>Assess tumor size and check for spread to new areas                                                                         |
| Every 3 months                                    | Complete blood count (CBC) and metabolic panel<br>Monitor for treatment-related bone marrow effects, kidney and liver function                                |
| Once (and as family members are identified)       | Germline genetic counseling follow-up<br>Discuss implications of BRCA1 germline mutation for first-degree relatives                                           |
| One-time recommendation for siblings and children | BRCA1/2 genetic testing for family members<br>First-degree relatives have a 50% chance of carrying the same BRCA1 mutation; testing allows preventive options |

### ⚠ When to Call Your Doctor

Contact your care team right away if you experience:

- Severe or new abdominal pain or bloating
- Difficulty breathing or new shortness of breath
- Unusual fatigue that does not improve with rest
- New pelvic pressure or back pain

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- Unexplained bruising or prolonged bleeding (platelet count may be low on therapy)

**Questions?** Your oncology nurse navigator or oncologist's on-call line

## For Your Family

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You carry a pathogenic BRCA1 germline mutation, meaning this change was present from birth and can be passed to your children. Each of your first-degree relatives (siblings, children, parents) has a 50% chance of carrying the same mutation. BRCA1 carriers have significantly elevated lifetime risk for ovarian cancer (39-44%) and breast cancer (57-65%). Preventive options exist, including risk-reducing salpingo-oophorectomy and enhanced breast cancer screening.

### Recommended Steps for Family Members

- Share this result with your first-degree relatives (siblings, children, parents)
- Encourage family members to discuss BRCA1/2 genetic testing with their physicians
- Children of BRCA1 carriers can be tested as adults for informed family planning

**Genetic Counseling Recommended:** A genetic counselor can help explain what these results mean for your family members and discuss testing options.

## Resources & Support

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You don't have to face this alone. These resources can help:



### Ovarian Cancer Research Alliance (OCRA)

Leading ovarian cancer advocacy organization providing clinical trial matching, financial support, and patient community connections.

📞 1-866-399-6262 | 🌐 <https://ocrahope.org>



### FORCE — Facing Our Risk of Cancer Empowered

Specialized support for BRCA mutation carriers and hereditary cancer — peer support, expert webinars, and family communication guides.

📞 1-866-288-7475 | 🌐 <https://www.facingourrisk.org>



### CancerCare Financial Assistance

Copay assistance for PARP inhibitors (olaparib/niraparib) and other targeted therapies. Includes navigation support for insurance prior authorization.

📞 1-800-813-4673 | 🌐 <https://www.cancercare.org>

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